LABETALOL

Pregnancy: first-line

Keep patient supine for IV

CHEMICAL NAME 2-hydroxy-5-{1-hydroxy-2-[(1-methyl-3-phenylpropyl)amino]ethyl}benzamide (hydrochloride)

CLASS Combined α1- and non-selective β-adrenoceptor blocker

TRADE NAMES Trandate · Normodyne · Labetalol HCl injection

THERAPEUTIC USES / INDICATIONS

- Hypertension in pregnancy and pre-eclampsia (first-line oral and IV).
- Hypertensive emergencies.
- Hypertension with angina.
- Controlled hypotension in surgery.

ADMINISTRATION

- **PO** with food, 2–3 times daily.
- **IV bolus:** 20 mg over 2 min, repeat; or **infusion**.
- Patient lying down during and 3 h after IV (postural hypotension).

DOSAGES

- **PO:** 100 mg twice daily, increase to 200–400 mg twice daily (max 2.4 g/day).
- **Severe hypertension in pregnancy (IV):** 20 mg, then 40 mg, then 80 mg every 10 min (max 300 mg); or infusion 20 mg/h doubling every 30 min (max 160 mg/h).
- **Hypertensive emergency:** 1–2 mg/min infusion.

PHARMACOKINETICS

- **Onset:** PO 20 min–2 h; IV 2–5 min. **Duration:** PO 8–12 h; IV 2–4 h.
- First-pass metabolism; crosses placenta.
- Excreted in urine/bile. **t½:** 6–8 h.

PHARMACODYNAMICS

- **α1 blockade** → vasodilation; **β blockade** prevents reflex tachycardia (β:α ratio ~3:1 PO, 7:1 IV).
- Lowers BP without reducing uteroplacental blood flow significantly.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Postural hypotension, dizziness.
- Fatigue, headache.
- Scalp tingling, nasal congestion.

ADVERSE EFFECTS (SERIOUS)

- Hepatotoxicity (rare, severe).
- Bronchospasm.
- Bradycardia, heart block.
- Neonatal bradycardia and hypoglycaemia if given near delivery.

CONTRAINDICATIONS & CAUTIONS

- Asthma or bronchospasm.
- 2nd/3rd-degree heart block, cardiogenic shock, severe bradycardia.
- Uncontrolled heart failure.
- **Caution:** liver disease, diabetes.

KEY INTERACTIONS

- **Verapamil, diltiazem:** heart block.
- **Halothane/volatile anaesthetics:** hypotension.
- **TCAs:** tremor.
- **Insulin:** masked hypoglycaemia.

NURSING MANAGEMENT

- IV: BP and HR every 5–15 min; patient supine; target BP per protocol (e.g. <150/100 in pregnancy).
- Continuous fetal monitoring in pregnancy.
- Inform neonatal team (monitor baby's glucose and HR).
- Check LFTs if symptoms of liver injury.
- Assist mobilisation after IV dose.

PATIENT EDUCATION

- Rise slowly; sit if dizzy.
- Report yellow skin, dark urine, wheeze.
- Keep antenatal BP checks.
- Scalp tingling is harmless and temporary.

KEY POINTS

- α + β blocker; first-line in pregnancy hypertension.
- Patient supine during IV.
- IV 20 → 40 → 80 mg for severe BP.
- Avoid in asthma.

SOURCES CHECKED **[310]** Labetalol 100mg/20ml solution for injection/infusion ampoules (5mg/ml) (UK SmPC) **[311]** NICE NG133 **[312]** Labetalol Hydrochloride, IV (US label)

Full references p. 432–457